

LECTURES 10-11

Example 0.1 (Poisson distribution with parameter λ). On $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ define probability measure

$$\mu(B) = \sum_{k \in B \cap \{0, 1, 2, \dots\}} \frac{\lambda^k e^{-\lambda}}{k!}, \quad B \in \mathcal{B}_{\mathbb{R}}.$$

Then μ defines a probability measure on $\mathcal{B}_{\mathbb{R}}$ and is called Poisson distribution with parameter λ . Corresponding distribution function is given by

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \sum_{k=0}^{\lfloor x \rfloor} \frac{\lambda^k e^{-\lambda}}{k!}, & x \geq 0 \end{cases}$$

Here observe that, we have defined directly the distribution and distribution function without starting with (Poisson) random variable opposed to what we have done in the previous examples.

Question: Is F is indeed a distribution function? From our definition of distribution function, F is a distribution function if there exists a random variable X such that $F(x) = P(X \leq x)$ for all $x \in \mathbb{R}$.

So we need to construct a random variable X satisfying $F(x) = P(X \leq x)$ for all $x \in \mathbb{R}$.

I will give one such construction. Observe that X should take values from $\{0, 1, 2, \dots\} = \{0\} \cup \mathbb{N}$. So take $\Omega = \{0\} \cup \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\Omega)$ and

$$P(A) = \sum_{k \in A} \frac{\lambda^k e^{-\lambda}}{k!}, \quad A \subseteq \Omega.$$

On this probability space, define $X : \Omega \rightarrow \mathbb{R}$ by $X(\omega) = \omega$. Then X is a random variable and (exercise)

$$P(X \leq x) = \sum_{k=0}^{\lfloor x \rfloor} \frac{\lambda^k e^{-\lambda}}{k!} = F(x), \quad x \geq 0$$

and $P(X \leq x) = 0 = F(x), x < 0$. i.e., F is the distribution function of X .

Example 0.2 (Geometric distribution) On $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ define probability measure satisfying

$$\mu(\{k\}) = p(1-p)^{k-1}, \quad k = 1, 2, \dots,$$

$0 < p < 1$. Then μ defines a probability measure on $\mathcal{B}_{\mathbb{R}}$ and is called geometric distribution with parameter p .

Student may add the details as in the previous example.

Example 0.3 (Uniform distribution on $[0, 1]$) The distribution function is given by

$$F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ x & \text{if } 0 < x \leq 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

is called the Uniform $[0, 1]$ distribution function.

As discussed above, first one need to check that the above indeed is a distribution function, i.e. we need to get(construct) a random variable X such that $P(X \leq x) = F(x), x \in \mathbb{R}$.

To this end, first we describe a probability space. On $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$, define the probability measure μ such that

$$\mu(B) = l(B \cap [0, 1])$$

when the Borel set B is an interval, where $l(B \cap [0, 1])$ denote the length of the interval $B \cap [0, 1]$ if it is non empty. The probability measure μ is called the uniform distribution on $[0, 1]$. Now take $(\Omega, \mathcal{F}, P)$ as $(\mathbb{R}, \mathcal{B}_{\mathbb{R}}, \mu)$ and define $X : \Omega \rightarrow \mathbb{R}$ as $X(\omega) = \omega$, the identity function. Then X is a random variable (exercise) and

$$P(X \leq x) = P((-\infty, x]) = l((-\infty, x] \cap [0, 1]) = F(x), x \in \mathbb{R}$$

Example 0.4 (Normal distribution with parameters m, σ)

The distribution function $F : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$F(x) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^x e^{-\frac{(y-m)^2}{2\sigma^2}} dy.$$

is called normal distribution function with parameters m, σ . Again to see that F is indeed a distribution function, I will give another useful construction of the 'normal' random variable. Let U be a uniform $[0, 1]$ random

variable defined on $(\Omega, \mathcal{F}, P)$. We can see that (exercise) F is strictly increasing and continuous with $0 < F(x) < 1$ for all $x \in \mathbb{R}$.

Define $X = F^{-1} \circ U$. Then (exercise) X is a random variable on $(\Omega, \mathcal{F}, P)$.

$$\begin{aligned} P(X \leq x) &= P(F^{-1} \circ U \leq x) \\ &= P(U \leq F(x)) = F(x), \text{ since } 0 < F(x) < 1. \end{aligned}$$

Before proceeding to the next example, I will give you the sketch of the original derivation of normal distribution due to Gauss. Question is the following: Given measurements (or observations) which are independent, is it possible to derive the distribution of the error, i.e. difference between measurement and the true value? Gauss postulated the following: most probable value for the true value x given the measurements $x_1, x_2, \dots, x_n$ is its average i.e.

$$\bar{x} = \frac{1}{n} \sum_{k=1}^n x_k.$$

Using the above postulate, Gauss derived the distribution of the error. Let $\varphi(\cdot)$ denote the 'density' of the error distribution which is assumed to be symmetric. To compute $\varphi(\cdot)$ Gauss (year 1809) used method of likelihood estimation (modern language). Consider the so called likelihood of x given the data

$$L(x; x_1, x_2, \dots, x_n) = \varphi(x_1 - x) \cdots \varphi(x_n - x).$$

According to Gauss' postulate, the value of x which maximizes $L(\cdot; x_1, x_2, \dots, x_n)$ is $\bar{x}$. let us compute the maximum.

Set $\psi(\theta) = \frac{\varphi'(\theta)}{\varphi(\theta)}$ and consider the log likelihood (easy to compute). Now Gauss' postulate implies

$$\sum_{k=1}^n \psi(x_k - \bar{x}) = 0. \quad \left(\text{hint : use } \frac{d}{dx} \ln L(x) = 0 \text{ at } \bar{x} \right)$$

Since $\sum_{k=1}^n (x_k - \bar{x}) = 0$, we have the following relation

$$\sum_{k=1}^n \psi(\theta_k) = 0, \text{ wherever } \sum_{k=1}^n \theta_k = 0. \quad (*)$$

Also $\varphi(\theta) = \varphi(-\theta)$ implies

$$\psi(\theta) = -\psi(-\theta). \quad (**)$$

In particular, take $n = 3$ in (*), and use (**), we get

$$\psi(\theta_1 + \theta_2) = \psi(\theta_1) + \psi(\theta_2).$$

Now assuming $\psi(\cdot)$ continuous, it follows that

$$\psi(\theta) = k\theta,$$

for some constant k . hence

$$\varphi'(\theta) = k\theta \varphi(\theta).$$

Solve this ODE using separation of variables, we get

$$\varphi(\theta) = ce^{k\theta^2}, \theta \in \mathbb{R}.$$

Since $\int \varphi(\theta)d\theta = 1$, it follows that $k < 0$, say $k = -h^2$ and $c = \frac{h}{\sqrt{\pi}}$, i.e. the Normal distribution.

Remark 0.1 *if one replace Gauss' postulate by "most probable value as median of the observation", what happens? We will see this later.*

Example 0.5 *(Exponential distribution with parameter $\lambda > 0$)*
The distribution function given by

$$F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 1 - e^{-\lambda x} & \text{if } x > 0 \end{cases}$$

is called exponential distribution function with parameter λ . Details of this example is left as an exercise.

Remark 0.2 *If $F : \mathbb{R} \rightarrow \mathbb{R}$ satisfies the following*

(1)

$$\lim_{x \rightarrow -\infty} F(x) = 0, \quad \lim_{x \rightarrow \infty} F(x) = 1,$$

(2) *F is nondecreasing and right continuous,*

then we can show that there exists a probability space $(\Omega, \mathcal{F}, P)$ and a random variable X on it such that

$$P(X \leq x) = F(x), \quad x \in \mathbb{R}.$$

In fact, we have seen this in the above examples. We have seen three methods to construct probability space and random variable on it satisfying $P(X \leq x) = F(x)$, $x \in \mathbb{R}$.

Method I: This method is for the 'discrete' F satisfying (1) and (2), i.e. F 'increases' only at jumps and hence all jumps add upto 1. Examples of such F 's are Bernoulli, Binomial, Poisson, Geometric etc. The precise definition of 'discrete' F is given in the subsection on 'classification of random variables'.

Here one take $\Omega = D = \{x_i | i \in I\}$, the set of discontinuities of F (here I is countable) and $\mathcal{F} = \mathcal{P}(\Omega)$ and P is defined by

$$P(\{x_i\}) = F(x_i) - F(x_i-), \quad i \in I,$$

i.e. $P(\{x_i\})$ is the jump size at x_i . Now define $X : \Omega \rightarrow \mathbb{R}$ as $X(\omega) = \omega$. Then

$$P\{X \leq x\} = \sum_{i: x_i \leq x} P\{X = x_i\} = \sum_{i: x_i \leq x} P\{x_i\} = F(x).$$

(Instruction: Student should carefully look at how each equality follows)

Method II: When F is (strictly) increasing and continuous, one can use the following. Let U be a uniform $[0, 1)$ random variable on a probability space $(\Omega, \mathcal{F}, P)$. Then define $X = F^{-1} \circ U$. Now as explained in the example of Normal distribution that

$$P(X \leq x) = P(U \leq F(x)) = F(x).$$

Method III: This method is very general and works for any F satisfying (1) and (2). Method relay on defining a probability measure P on $(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ such that

$$P((-\infty, x]) = F(x), \quad x \in \mathbb{R}.$$

(Here note that P is nothing but the distribution μ corresponding to F). Now define $X : \Omega \rightarrow \mathbb{R}$ by $X(\omega) = \omega$. Then

$$P(X \leq x) = P((-\infty, x]) = F(x), \quad x \in \mathbb{R}.$$

Thus we have the following equivalent definition of distribution function on $\mathbb{R}$.

Definition 5.3 (distribution function)

A function $F : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

(1)

$$\lim_{x \rightarrow -\infty} F(x) = 0, \quad \lim_{x \rightarrow \infty} F(x) = 1,$$

(2) F is nondecreasing and right continuous

is said to be a distribution function on $\mathbb{R}$.

A Classification of random variables. Random variables can be classified using distribution functions according to their 'continuity properties'.

Definition 5.4: A random variable X with distribution function $F : \mathbb{R} \rightarrow \mathbb{R}$ is said to be a discrete random variable if

$$\sum_{x \in D} (F(x) - F(x-)) = 1,$$

where D is the set of discontinuities of F .

Here observe that the 'discrete' distribution F 'exhaust' all the probability masses through the jumps.

Let X be the random variable which is 'degenerate at 0, i.e. $X \equiv 0$, the its distribution function is given by

$$H_0(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$

and is called the Heaviside function. The distribution corresponding to $H_0(\cdot)$ is (exercise)

$$\delta_0(B) = \begin{cases} 0 & \text{if } 0 \notin B \\ 1 & \text{if } 0 \in B \end{cases}$$

and δ_0 is called the 'Dirac delta' probability measure. If X is degenerate at a , then its distribution function is $H_a(x) = H_0(x - a)$, $x \in \mathbb{R}$ and the corresponding distribution is δ_a .

Lemma 0.1 *If F is the distribution of a discrete random variable X , then it is of the form*

$$F(x) = \sum_{i: x_i \in D} p_i H_0(x - x_i), \quad x \in \mathbb{R},$$

where H_0 denote the Heaviside function and $p_i = P(X = x_i) = F(x_i) - F(x_i-)$ and D is the set of discontinuities of F .

Proof: Let $D = \{x_i | i \in I\}$ where the index set I is countable and X be a random variable with distribution F . Then it follows that $P(X = x) = 0$ for all $x \notin D$. For $x \in \mathbb{R}$, let $I_x = \{i \in I | x_i \leq x\}$. Then observe that $i \in I_x \Leftrightarrow H_0(x - x_i) = 1$ and hence $i \notin I_x \Leftrightarrow H_0(x - x_i) = 0$. Therefore,

$$\begin{aligned} F(x) &= P(X \leq x) \\ &= \sum_{i \in I_x} P(X = x_i) = \sum_{i \in I_x} p_i \times 1 + \sum_{i \notin I_x} p_i \times 0 \\ &= \sum_{i \in I} p_i H_0(x - x_i). \end{aligned}$$

□

What the above result telling us is any discrete random variable with distribution function F is a convex combination of random variables which are degenerate at the discontinuities of F .

The distributions in Examples 1, 2, 3 of Lecturenote 9 and 0.1 corresponds to discrete random variables.

Definition 5.5 A random variable X with distribution function $F : \mathbb{R} \rightarrow \mathbb{R}$ which is continuous is said to be random variable with continuous distribution and is in short called by the name continuous random variable.

The distributions given in Examples 0.4, 0.3, 0.5 corresponds to continuous random variables.

Definition 5.6 (Probability mass function)

Let X be a discrete random variable with distribution function $F : \mathbb{R} \rightarrow \mathbb{R}$.

Define $f : \mathbb{R} \rightarrow \mathbb{R}$ as follows:

$$f(x) = F(x) - F(x-)$$

Then f is called the probability mass function (pmf) of X .

For example, the pmf of the discrete random variable of Example ?? is given by

$$f(x) = \begin{cases} \frac{1}{4} & \text{if } x = 0, 2 \\ \frac{1}{2} & \text{if } x = 1 \\ 0 & \text{otherwise.} \end{cases}$$

It is left as an exercise for the student to write down the pmf of random variables in Examples 2, 3 of Lecturenote 9 and 0.1.

The pmf of a continuous random variable is the zero function. Hence the notion of pmf is useless for continuous random variables.

Definition 5.7 (Probability density function)

A continuous random variable X with distribution function $F : \mathbb{R} \rightarrow \mathbb{R}$ is said to have a probability density function(pdf) if there exists a function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$F(x) = \int_{-\infty}^x f(y)dy \quad \forall \quad x \in \mathbb{R}$$

If $f : \mathbb{R} \rightarrow \mathbb{R}$ exists, then it is called the pdf of X .

A continuous random variable with a pdf is called **absolutely continuous** random variable.

It is easy to see that if F is differentiable every where and the derivative denoted by F' is a continuous function, then the corresponding random variable X has a pdf and is given by $f = F'$. This is not a necessary condition.

Example 0.6 Define $F : \mathbb{R} \rightarrow \mathbb{R}$ as follows.

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } 0 \leq x < \frac{1}{2} \\ \frac{1}{2} & \text{if } \frac{1}{2} \leq x < 1 \\ x - \frac{1}{2} & \text{if } 1 \leq x < \frac{3}{2} \\ 1 & \text{if } x \geq \frac{3}{2}. \end{cases}$$

Student can verify that F is a distribution function as it satisfies the conditions (1) and (2) given in the definition 5.3. Also verify that it corresponds to the random variable given by the random experiment of picking a point 'at random' from $[0, \frac{1}{2}] \cup [1, \frac{3}{2}]$.

F is not differentiable at $x = \frac{1}{2}, 1$. Verify that the function

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } 0 \leq x < \frac{1}{2} \\ 0 & \text{if } \frac{1}{2} \leq x < 1 \\ 1 & \text{if } 1 \leq x < \frac{3}{2} \\ 0 & \text{if } x \geq \frac{3}{2}. \end{cases}$$

is the pdf of F (exercise).

Consider the distribution function

$$\begin{aligned} F(x) &= 0 & \text{if } x < 0 \\ &= \frac{1}{4} & \text{if } 0 \leq x < 1 \\ &= \frac{3}{4} & \text{if } 1 \leq x < 2 \\ &= 1 & \text{if } x \geq 2. \end{aligned}$$

Note that F is differentiable except at the points 0, 1, and 2 and $F'(x) = 0$ for $x \neq 0, 1, 2$. Here the pdf doesn't exist. Compare this with the distribution function in Example 0.6.

Distribution function of transformation of random variables: In this subsection, we will see how one can write down the distribution function of $Y = \varphi \circ X$ in terms of the distribution of X where $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is Borel measurable. Note that it is not possible to give an explicit formula but in some cases one will be able to do that. Here my plan is to give a general recipe and will illustrate it through some examples. I will give as an example, one special class of transformations. Though it is possible to give an explicit formula for many other cases, I will not do it instead take some examples and show you how to use it.

We use μ_Z, F_Z to denote the distribution and distribution function of a random variable Z from now onwards.

General Recipe: The distribution of Y is given by

$$\begin{aligned}\mu_Y(B) &= P(Y \in B) \\ &= P(\varphi(X) \in B) \\ &= P(X \in \varphi^{-1}(B)) \\ &= \mu_X(\varphi^{-1}(B)), \quad B \in \mathcal{B}_{\mathbb{R}}.\end{aligned}$$

Hence by taking $B = (-\infty, y]$, $y \in \mathbb{R}$, we get following:

$$F_Y(y) = \mu_X(\varphi^{-1}(-\infty, y]), \quad y \in \mathbb{R}.$$

Hence to compute the distribution function Y in terms of the distribution function of X , one need to identify the set $\varphi^{-1}(-\infty, y]$. This I will illustrate in the next example.

Lemma 0.2 *Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function which is non decreasing. Then $\varphi^{-1}(-\infty, y] = (-\infty, \sup \varphi^{-1}(y)]$. This implies that*

$$F_Y(y) = F_X(\sup \varphi^{-1}(y)), \quad y \in \mathbb{R}.$$

In particular, if φ is strictly increasing, then $F_Y(y) = F_X(\varphi^{-1}(y))$.

Proof: (Reading exercise) Now we will see the proof of $\varphi^{-1}(-\infty, y] = (-\infty, \sup \varphi^{-1}(y)]$.

$$\begin{aligned}x \in \varphi^{-1}(-\infty, y] &\Rightarrow \varphi(x) \in (-\infty, y] \\ &\Rightarrow \varphi(x) < y \text{ or } \varphi(x) = y \\ &\Rightarrow x < z \text{ for all } z \in \varphi^{-1}(y) \text{ or } \varphi(x) = y \\ &\Rightarrow x \in (-\infty, \sup \varphi^{-1}(y)].\end{aligned}$$

The proof of statement $\varphi(x) < y \Rightarrow x < z$ for all $z \in \varphi^{-1}(y)$ is as follows. Suppose there exists some $z \in \varphi^{-1}(y)$ such that $x \geq z$, then $\varphi(x) \geq \varphi(z) = y$, a contradiction to $\varphi(x) < y$.

Now we prove the reverse inclusion. Suppose $x \leq \sup \varphi^{-1}(y)$. Then either (I) : $x < \sup \varphi^{-1}(y) \Rightarrow x \leq z$ for some $z \in \varphi^{-1}(y)$ or (II) : $x = \sup \varphi^{-1}(y) \Rightarrow x \geq z$ for all $z \in \varphi^{-1}(y)$ and there exists a sequence $z_n \in \varphi^{-1}(y)$ satisfying $z_n \rightarrow x$.

Now

$$\begin{aligned} \text{(I)} \quad &\Rightarrow \varphi(x) \leq y \\ &\Rightarrow x \in \varphi^{-1}((-\infty, y]). \end{aligned}$$

$$\begin{aligned} \text{(II)} \quad &\Rightarrow \varphi(x) = \lim_{n \rightarrow \infty} \varphi(z_n) = y \text{ (using continuity of } \varphi) \\ &\Rightarrow \varphi(x) = y \\ &\Rightarrow x \in \varphi^{-1}(y) \subseteq \varphi^{-1}((-\infty, y]). \end{aligned}$$

This completes the proof of the reverse inclusion. Hence the proof is complete.

I will give an example to illustrate the above.

Example 0.7 Let $\varphi(x) = x^+, x \in \mathbb{R}$. Then

$$\varphi^{-1}(y), \sup \varphi^{-1}(y) = \begin{cases} \emptyset, & -\infty \text{ if } y < 0 \\ (-\infty, 0], & 0 \text{ if } y = 0 \\ y, & y \text{ if } y > 0. \end{cases}$$

Lemma 0.3 Let X be a random variable with pdf f such that $f \equiv 0$ outside an interval I and $\varphi : I \rightarrow \mathbb{R}$ is a strictly monotone and differentiable function. Then pdf g of $\varphi \circ X$ exists and is given by

$$g(y) = f(\varphi^{-1}(y))|(\varphi^{-1})'(y)|, y \in \varphi(I), \quad g(y) = 0 \text{ elsewhere.}$$

Proof: (Reading exercise) When φ is strictly increasing, and let a, b denote respectively the left and right endpoints of I . For $y \in \varphi(I)$,

$$\begin{aligned} F_Y(y) &= P\{\varphi \circ X \leq y\} \\ &= P\{X \leq \varphi^{-1}(y)\} \\ &= F_X(\varphi^{-1}(y)). \end{aligned}$$

Hence

$$F_Y(y) = \begin{cases} 0 & \text{if } y < \varphi(a) \\ F_X(\varphi^{-1}(y)) & \text{if } \varphi(a) \leq y < \varphi(b) \\ 1 & \text{if } y \geq \varphi(b). \end{cases}$$

Now for $\varphi(a) \leq y < \varphi(b)$,

$$\begin{aligned} F_Y(y) &= F_X(\varphi^{-1}(y)) \\ &= \int_{-\infty}^{\varphi^{-1}(y)} f(t) dt = \int_a^{\varphi^{-1}(y)} f(t) dt \\ (\text{put } t = \varphi^{-1}(u)) &= \int_{\varphi(a)}^y f(\varphi^{-1}(u)) (\varphi^{-1})'(u) du \\ &= \int_{-\infty}^y g(u) du, \end{aligned}$$

where

$$g(y) = \begin{cases} f(\varphi^{-1}(y)) (\varphi^{-1})'(y) & \text{if } \varphi(a) \leq y < \varphi(b) \\ 0 & \text{otherwise} \end{cases}$$

For $y \notin [\varphi(a), \varphi(b))$, it is easy to see that

$$F_Y(y) = \int_{-\infty}^y g(u) du.$$

This completes the proof. $\square$

Example 0.8 Let X be Uniform $(-\frac{\pi}{2}, \frac{\pi}{2})$ random variable and $Y = \tan X$.

Here note $\varphi(x) = \tan x$, $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$ is strictly increasing, differentiable and $\varphi(I) = \mathbb{R}$. Hence using Lemma 0.3, it follows that Y has a pdf g given by

$$\begin{aligned} g(y) &= f(\tan^{-1} y) \left| \frac{d}{dy} \tan^{-1} y \right|, \quad y \in \mathbb{R} \\ &= \frac{1}{\pi} \frac{1}{1+y^2}, \quad y \in \mathbb{R}, \end{aligned}$$

where $f(x) = \frac{1}{\pi}$, $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$ and $= 0$ otherwise. Hence Y is a Cauchy random variable.

Example 0.9 Let $\varphi(x) = x^2 + 1$. (Prototype for convex functions). Then

$$\varphi^{-1}(-\infty, y] = \begin{cases} \emptyset & \text{if } y < 1 \\ \{0\} & \text{if } y = 1 \\ [-\sqrt{y-1}, \sqrt{y-1}] & \text{if } y > 1. \end{cases}$$

Hence

$$F_Y(y) = \mu_X([- \sqrt{y-1}, \sqrt{y-1}]) = F_X(\sqrt{y-1}) - F_X(\sqrt{y-1}-), y > 1,$$

$$F_Y(1) = \mu_X(\{0\}), \text{ and } F_Y(y) = 0, y < 1.$$

Warning : Don't use Lemma 5.3 in this case as φ is not monotone.

Example 0.10 φ be the Heaviside function, i.e. $\varphi(x) = 0$ if $x < 0$ and $= 1$ if $x \geq 0$. (Prototype for φ which is piece-wise continuous) Then

$$\varphi^{-1}(-\infty, y] = \begin{cases} \emptyset & \text{if } y < 0 \\ (-\infty, 0) & \text{if } 0 \leq y < 1 \\ \mathbb{R} & \text{if } y \geq 1. \end{cases}$$

Hence

$$F_Y(y) = \begin{cases} 0 & \text{if } y < 0 \\ F_X(0-) & \text{if } 0 \leq y < 1 \\ 1 & \text{if } y \geq 1. \end{cases}$$

Example 0.11 Let X be a normal with parameters 0 and σ and $Y = X^2$, i.e. $\varphi(x) = x^2, Y = \varphi \circ X$. Then

$$\varphi^{-1}(-\infty, y] = \begin{cases} \emptyset & \text{if } y < 0 \\ [-\sqrt{y}, \sqrt{y}] & \text{if } y \geq 0. \end{cases}$$

Hence

$$F_Y(y) = F_X(\sqrt{y}) - F_X(-\sqrt{y}), y \geq 0.$$

Hence the pdf g of Y is given by

$$g(y) = \frac{1}{2\sqrt{y}}(f(\sqrt{y}) + f(-\sqrt{y})), y \geq 0,$$

where f is the pdf of X and is given by

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}, x \in \mathbb{R}.$$

Hence

$$g(y) = \frac{1}{\sqrt{2\pi y}\sigma} e^{-\frac{y}{2\sigma^2}}, y > 0,$$

i.e., Y follows Gamma distribution with parameters $\frac{1}{2}$ and $\frac{1}{2\sigma^2}$.

Recall that a random variable X follows gamma with parameters (α, λ) if its pdf is given by

$$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, x > 0, \quad g(y) = 0, \quad x \leq 0,$$

where

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx, \alpha > 0.$$

Example 0.12 Let X be uniform $(0, 1)$ and $Y = \max\{\frac{1}{2}, X\}$. Then $\varphi(x) = \max\{\frac{1}{2}, x\}$, $x \in \mathbb{R}$ an increasing function. Here

$$\varphi^{-1}(y) = \begin{cases} \emptyset & \text{if } y < \frac{1}{2} \\ (-\infty, \frac{1}{2}] & \text{if } y = \frac{1}{2} \\ \{y\} & \text{if } y > \frac{1}{2}. \end{cases}$$

$$F_Y(y) = F_X(\sup \varphi^{-1}(y)) = \begin{cases} 0 & \text{if } y < \frac{1}{2} \\ F_X(\frac{1}{2}) & \text{if } y = \frac{1}{2} \\ F_X(y) & \text{if } y > \frac{1}{2}. \end{cases}$$

Hence

$$F_y(y) = \begin{cases} 0 & \text{if } y < \frac{1}{2} \\ \frac{1}{2} & \text{if } y = \frac{1}{2} \\ y & \text{if } \frac{1}{2} < y < 1 \\ 1 & \text{if } y \geq 1. \end{cases}$$